

1. Executive Overview

NovaTech Solutions is a fictional technology-products company created for testing a business-analysis and RAG workflow.

Objective: understand revenue, profitability, product contribution, regional performance, customer segments, discount behavior, and time-based sales patterns.

Scope: four regions, five products, three customer segments, multiple sales channels and representatives, order quantities, prices, discounts, revenue, cost, and profit.

Testing rule: numerical conclusions should be calculated from the accompanying Excel dataset.

2. Company & Market Context

NovaTech sells Business Laptops, Monitors, Wi-Fi Routers, Keyboard & Mouse products, and Cloud Suite subscriptions.

Orders are generated across North, South, East, and West regions.

Customers are grouped into Enterprise, SMB, and Consumer segments.

Business questions include regional revenue, product profitability, order value, customer performance, and discount behavior.

3. Revenue & Profitability

Revenue represents net sales after the recorded discount.

Cost represents recorded product cost. Profit is revenue minus cost.

Profit Margin is profit divided by revenue.

Suggested calculations: total revenue, total profit, units, orders, average order value, average discount, margin, and revenue/profit by dimension.

Revenue and profit should always be compared together.

4. Product Portfolio Analysis

The portfolio contains Business Laptop, Monitor, Wi-Fi Router, Keyboard & Mouse, and Cloud Suite.

Identify the product with the highest revenue, highest profit, highest quantity, and highest average selling value.

Compare product margins rather than judging products only by revenue.

Order-level data can be used to investigate cross-selling between hardware, accessories, and Cloud Suite.

5. Regional Performance

NovaTech operates across North, South, East, and West.

Calculate revenue, profit, quantity, order count, average order value, and margin for each region.

Investigate whether regional differences are associated with order volume, product mix, pricing, or discounts.

Combine region with product and customer segment for deeper analysis.

6. Customer Segment Analysis

Customers are grouped into Enterprise, SMB, and Consumer segments.

Calculate revenue, profit, quantity, order count, average order value, average discount, and margin for each segment.

Compare order size and frequency across segments.

Do not assume which segment performs best before calculating the actual dataset metrics.

7. Discount & Pricing Analysis

Discounting can affect both revenue and profitability.

Compare revenue, quantity, and margin across discount levels.

Identify products and regions receiving larger discounts.

An association between discount and quantity does not prove that discounting caused higher demand.

8. Time-Series & Demand Analysis

Order dates can be grouped by month to examine sales activity over time.

Calculate monthly revenue, profit, quantity, and order count.

Compare periods and identify unusually high or low activity.

Forecasts are estimates and should be validated against historical behavior.

9. Data Quality Checklist

Check missing values, duplicate Order_ID values, invalid dates, negative quantities, invalid prices, unexpected discounts, and inconsistent category names.

Recalculate revenue and profit from underlying fields and validate stored values.

Every reported number should be traceable to the uploaded dataset.

Clearly define units, date ranges, and comparison populations.

10. Management Questions & Next Analysis

Priority questions: What are total revenue and profit? Which region contributes most revenue? Which product contributes most profit?

Also examine customer segment order value, discount-versus-margin behavior, and monthly sales trends.

A dashboard can include revenue, profit, units, orders, margin, regional comparison, product contribution, segment performance, and monthly trends.

Next step: upload the Excel dataset and ask natural-language questions so the AI assistant calculates the answers from the records.

This document intentionally contains exactly ten pages for a faster RAG-processing test.